

5. Precipitation

Std 9th

Sub – Geography

Q1 Identify the precipitation type with the help of the description given:

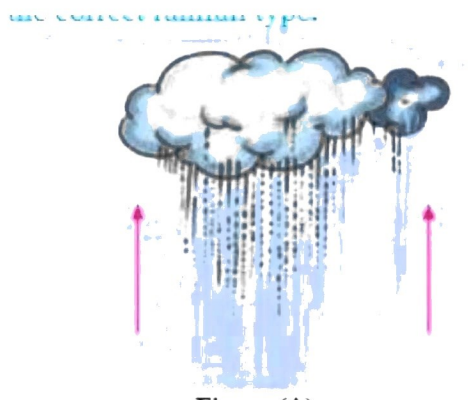
(a) It is the main source of the water that you use. Sometimes it is torrential and sometimes continuous. Most of the agriculture in India is dependent on it. - **Rainfall**

(b) It seems as if water droplets are floating in the atmosphere. In London, one cannot see the Sun till the afternoon during winters because of this phenomenon. - **Fog**

(c) It never precipitates like this in equatorial areas. Precipitation in the solid form sometimes causes damage to the crops. - **Hail**

(d) A white cotton like layer spreads on the earth's surface. Because of this form of precipitation, the State of Jammu and Kashmir has to change its capital in winters. In Maharashtra, it does not precipitate like this. - **Snow**

Q2 Look at the following pictures and identify the correct rainfall type.



Convectional rainfall



Orographic rainfall



Figure (C)

Cyclonic rainfall

Q 3. Look at the figures above and answer the following questions:

(1) In fig B, on which side of the mountain is it raining more?

Ans. The windward side is receiving more rainfall

(2) Shade the rain shadow region in fig B and name it.

Ans. Students to show leeward side in the picture.

(3) What is the difference between A and C?

Ans. 1. In figure A i.e. convectional rainfall the hot air rises upwards and then the air cools and begins to condense and due to continuous condensation rainfall occurs.

2. Hence rainfall is accompanied by rainfall lighting and thunder.

3. In figure C, i.e. cyclonic rainfall, air from surrounding regions comes towards the centre of the cyclone and starts moving upwards. As it rises, the temperature of the air reduces, condensation occurs and rainfall takes place.

(4) Stormy winds and floods are associated with which rainfall type?

Ans. Stormy winds and floods are associated with Cyclonic rainfall.

(5) What type of rainfall occurs in Singapore?

Ans. Cyclonic rainfall occurs in Singapore.

Q 4. Identify the odd man out :

(1) Orographic rainfall, acid rain, cyclonic rainfall, convectional rainfall - **acid rain**

(2) Snowfall, rainfall, hailstones, dew - **dew**

(3) Thermometer, rain gauge, anemometer, measuring jar - **measuring jar**

Q 5. Answer in brief:

(1) In what ways does precipitation occur on the earth?

Ans. Precipitation means waterfall in the solid or liquid state the clouds to the earth's surface. Snow, hailstorms, rainfall are the major forms of precipitation.

Snow: 1. When the temperature in the atmosphere falls below the freezing point, water vapour directly turns into snowflakes. This is called sublimation.

2. Precipitation in the form of solid particles of snow is known as snowfall.

4. As snow is in the solid form, it does not flow like water. When the snow melts, the region gets fresh water.

Hail: 1. When there is lot of heat on the earth's surface, the upward air flow blows at a greater speed. Because of this upward flow, the temperature of the air reduces and the condensation of the water vapour takes place and Dark clouds are formed.

2. Because of the upward movement of air, these water droplets go to a higher altitude.

3. Here, solidification of these droplets occur and hailstones are formed. As hailstones are heavy, they fall towards the ground,

crops may get destroyed and loss of life and property may also occur.

Rainfall: 1. We generally get water in the form of rainfall. The temperature of the air with water vapour reduces when it goes higher.

2. Condensation of the vapour occurs. Clouds form when the condensed water droplets and dust particles accumulate. The water droplets increase in size.

3. When they cannot float in the air anymore because of their weight, they come down as rainfall.

Fog, dew and frost : When the condensation or solidification of the water vapour in the atmosphere occurs near the earth's surface, then we can see fog, dew or frost.

(2) Comment on the rainfall occurring in the rain shadow area.

Ans. 1. Winds coming from lakes or seas are moisture-laden.

2. They are obstructed by the high mountain ranges coming in their way.

3. They start going upwards along the slope of the mountains, The temperature of these winds drop and condensation occurs and rainfall takes place.

4. Thus because of the obstruction of the mountains, this type of rainfall occurs.

5. The windward side of the mountains gets more rain; amount of vapour in the air reduces after crossing the mountain and the moisture holding capacity of the air increases.

(3) Which type of rainfall occurs in most of the world? Why?

Ans. 1. Orographic rainfall occurs in most of the parts in the world.

2. Convectional rainfall is regional in nature.

3. There is a certainty in the convectional rainfall occurring in the equatorial areas. Comparatively, the orographic and cyclonic rainfalls are less certain. And therefore, such areas are prone to very heavy rainfall, floods or droughts frequently.

(4) If condensation occurs closer to the earth's surface, what types of forms become visible?

Ans. When the condensation or solidification of the water vapour in the atmosphere occurs near the earth's surface, then we can see fog, dew or frost.

(5) What precautions should be taken while measuring rainfall?

Ans. 1. Rainfall is an important source of water on the earth and rainfall is formed because of changes in the temperature of the air with water vapour.

2. The instrument that is used to measure rainfall is called rain gauge. The funnel that is used in the rain gauge has a specific diameter.

3. The rain falling in this funnel is collected in a bottle fitted in the gauge.

4. The collected water is then measured with the help of measuring jar.

3. In areas of heavy rainfall, the reading of the rain is taken every three hours.

4. The measuring jar reads in millimeters. The gauge is kept in open ground on a 30cm high flat mound.

5. Hence, the rainwater is collected without any obstruction

Q 6. Distinguish between -

(1) Dew and frost

Ans.

Dew	frost
1. When moist laden air near the earth's surface comes into contact with very cold objects, condensation of vapour takes place into small water droplets called dew.	1. If the temperature of air is less than 0°C the water droplets stuck to the cold surface objects, freezes forming frost.
2. Water vapour condenses and forms droplets of water.	2. Water droplets stuck to cold surface turns to frozen water droplets.

3. Dew sticks to the cold object but does not freeze.	3. It sticks to the cold object and freeze.
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(2) Snow and hail

Ans.

Snow	Hail
1. Precipitation in the form of solid particles of snow is known as snow fall.	1. Precipitation in the form of frozen water droplets falling rapidly to the ground is known as hail.
2. The fall of temperature in atmosphere below the freezing point causes snow fall.	2. Extreme heat on the earth's surface initiates the process of hail formation.
3. Heavy accumulation of snow can collapse transportation and communication system of the area.	3. It destroys crops and causes loss of life and property.